Student Number:									

3 Complex Analysis 2024

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Final Exam

12 November 2024

Instructions:

- This question paper consists of **7 pages** (including this one).
- You have 120 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 60 points = 100% (63 points available).

Problem	1	2	3	4	5	6	Σ	Mark
Maximum Points	10	8	12	13	10	10	60	100%
Result								

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Problem 1

(10 points)

Find the set of all $z \in \mathbb{C}$ where the function $f(z) = (\operatorname{Re} z)^2$ is differentiable.

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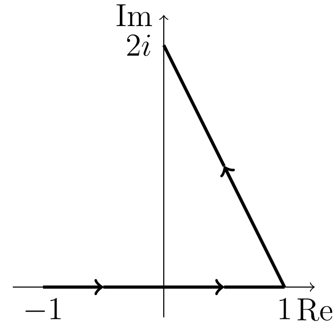
Problem 2

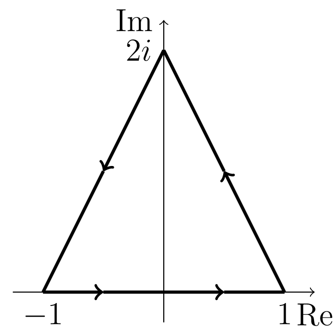
(8 points)

Calculate the following integrals.

(a) $\displaystyle\int_{\gamma_1} \frac{1}{(z-i)^2}\, dz$, where γ_1 is the indicated path:



(b) $\displaystyle\int_{\gamma_2} \frac{1}{(z-i)^2}\, dz$, where γ_2 is the indicated path:



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Problem 3

(12 points)

(a) Show that there exists no holomorphic function $f : \mathbb{C} \to \mathbb{C}$ with

$$\left| f'\left(\frac{1}{2}\right) \right| > 4 \max_{|z|=1} |f(z)|.$$

(b) Let $f : \mathbb{C} \to \mathbb{C}$ be a holomorphic function such that

$$f(z) \neq 0 \quad \text{if} \quad |z| = 1.$$

Put $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$. Assume that

$$\int_\gamma \frac{1}{f(z)} \, dz \neq 0.$$

Prove that f has at least one zero in the open unit disc $D(0; 1) = \{z \in \mathbb{C} : |z| < 1\}$.

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Problem 4

(13 points)

(a) Determine the singularities of the function $f(z) = \dfrac{1}{z^4 - 1}$ in $\mathbb{C}$ and their type. For poles, give the order. Justify your answer.

(b) Calculate the residue at each of the poles of f.

(c) Calculate P.V. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{x^4 - 1}\, dx$.

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Problem 5

(10 points)

(a) Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{n^n}{n!} z^{2n-1}$.

(b) Determine the Laurent series of the function $f(z) = \frac{e^{2z}}{(z+1)^2}$ about the point $z_0 = -1$. Give an explicit formula for the coefficients a_n, $n \in \mathbb{Z}$, of the Laurent series. Give the region of convergence of this series.

(c) Use the Laurent series in (b) to determine the residue of the function $f(z) = \frac{e^{2z}}{(z+1)^2}$ at $z_0 = -1$.

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Problem 6

(10 points)

How many zeros does the polynomial $p(z) = z^3 - 4z^2 - 5$ have in the set $\{z \in \mathbb{C} : |z| > 2\}$?